

Electric charge

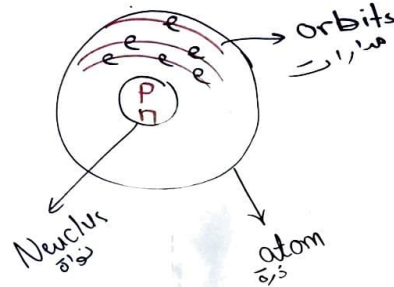
Fundamental charge $\rightarrow e = 1.6 \times 10^{-19} \text{ C}$

Charged particles:

$$\begin{aligned} \textcircled{1} \text{ electron} &\rightarrow \frac{q}{e} = -e & \left\{ \begin{array}{l} m_e = 9.11 \times 10^{-31} \text{ kg} \end{array} \right. \\ \textcircled{2} \text{ Proton} &\rightarrow \frac{q}{p} = +e & \left\{ \begin{array}{l} m_p = 1.67 \times 10^{-27} \text{ kg} \end{array} \right. \end{aligned}$$

$$\text{Neutron} \rightarrow \frac{q}{n} = 0 \quad \left\{ \begin{array}{l} m_n = 1.67 \times 10^{-27} \text{ kg} \end{array} \right.$$

متعاد الشحنة
Neutral



التيار الكهربائي
* Electric current:



موصل كهربائي
conductor

$$I = \frac{Q}{t}$$

Charge per unit time

$$I \rightarrow \text{C/s} \Rightarrow \text{Ampere (A)}$$

A number of electric charges passing through a conductor in 1 second.

Electric charge

Type of materials:

① Conductors:

Materials allow the current to pass, having free electrons
examples: ① Gold ② Silver ③ Copper ④ Aluminum

② Semiconductors:

② Insulators:

Materials prevent the current to pass, having no free electrons

examples: ① Wood
② Glass
③ Plastic
④ Papers - Ceramic etc.

↓
bonded electrons

س.
II

$$q = -3.2 \times 10^{-19} \text{ C} \quad \checkmark$$

$$q = 4.8 \times 10^{-19} \text{ C} \quad \checkmark$$

$$q = 3.2 \times 10^{-18} \text{ C} \quad \checkmark$$

$$q = 3.2 \times 10^{-20} \text{ C} \quad \times$$

Electric charge

Properties

- ① same charge $\rightarrow$ repel each other (repulsive force)
 opposite charge $\rightarrow$ attract each other (attractive force)
- ② Total charge in isolated system is Conserved
 Charge is not created nor destroyed
- ③ Charge is Quantized
 $e = 1.6 \times 10^{-19} \text{ C}$
 $\rightarrow$ charge can be integer multiple of e

$$q = n \times e$$

الشحنة الكلية = عدد الشحنات $\times$ الشحنة الأولية

$$n = \frac{q}{e}$$

$$m_t = n \times m$$

$$n = \frac{m_t}{m}$$

Physics II

$$1.6 \times 10^{-19} \text{ C}$$

$$I \rightarrow 1 \text{ C/s}$$

Electric charge

How many electrons in -1 C .

$$n = \frac{q}{e} = \frac{-1}{-1.6 \times 10^{-19}} = 6.25 \times 10^{18} \text{ electrons}$$
$$= \underline{625 \times 10^{16} \text{ electrons}}$$

Physics II

A group of protons having $320 \times 10^{-19} \text{ C}$. Find the mass of the group.

$$n = \frac{q_t}{e} = \frac{320 \times 10^{-19}}{1.6 \times 10^{-19}} = 200 \text{ protons}$$

$$m_t = n \times m_p = 200 \times 1.67 \times 10^{-27} = 334 \times 10^{-27} \text{ kg}$$

A group of proton having $250.5 \times 10^{-27} \text{ kg}$. Find the charge of the group.

Electric charge

How many electrons in -1 C .

$$n = \frac{q_t}{e} = \frac{-1}{-1.6 \times 10^{-19}} = 6.25 \times 10^{18} \text{ electron}$$
$$= \underline{625 \times 10^{16} \text{ electrons}}$$